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EE1060: Differential Equations and Transform Techniques

Lecture 25: Fourier Transform

Topics :

1. Properties of Fourier Transform
 2. Convergence of Fourier Transform
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So far

$$\text{FT of } x(t) : X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt \quad (\text{Analysis})$$

$$F(x(t)) = X(\omega)$$

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega \quad (\text{Synthesis})$$

→ existence

 $x(\omega)$ is finite

$$\int_{-\infty}^{\infty} |x(t)|^2 dt < \infty$$

→ Convergence

$$S_N(t) \rightarrow x(t)$$

Reconstruction

$$a) \int_{-\infty}^{\infty} |MSG| |S_N - x(t)|^2 dt$$

b) Pointwise Convergence.

Examples:-

$$\textcircled{1} \delta(t) \Leftrightarrow 1$$

$$\textcircled{2} 1 \Leftrightarrow ? \quad 2\pi \delta(\omega)$$

$$\textcircled{3} u(t) \Leftrightarrow ?$$

$$\int_{-\infty}^{\infty} x(t) \delta(t-\tau) dt = x(\tau)$$

Prop of FT:-

$$\textcircled{1} F(x_1(t) + x_2(t)) = F(x_1(t)) + F(x_2(t))$$

$$\textcircled{2} F(x(t-\tau)) = e^{-j\omega\tau} F(x(t))$$

$$\textcircled{3} F(x(at)) = \frac{1}{|a|} X\left(\frac{\omega}{a}\right)$$

$$\textcircled{4} F(\dot{x}(t)) = j\omega X(\omega)$$

$$\textcircled{5} \text{ if } x(t) \Leftrightarrow X(\omega)$$

$$\text{then } F(x(t)) = 2\pi X(-\omega)$$

↓
replace ω with $-\omega$ ↓
replace t with $-t$ Dirichlet Cond

→ Chap 4 of
asgood → distributions

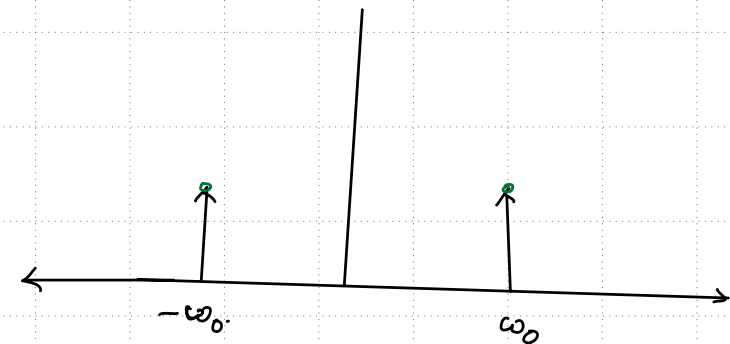
Examples

$$\begin{aligned}
 \textcircled{1} \quad F\{x(t) e^{-j\omega_0 t}\} &= \\
 &= \int_{-\infty}^{\infty} x(t) e^{-j\omega_0 t} e^{-j\omega t} dt \\
 &= \int_{-\infty}^{\infty} x(t) e^{+j(\omega + \omega_0) t} dt \\
 &= x(\omega + \omega_0)
 \end{aligned}$$

$$F\{e^{j\omega_0 t}\} = 2\pi \delta(\omega - \omega_0)$$

$$\begin{aligned}
 \textcircled{2} \quad F\{x(t) \cos \omega_0 t\} &= \\
 &= \frac{F\{x(t) e^{j\omega_0 t}\} + F\{x(t) e^{-j\omega_0 t}\}}{2} \\
 &= \frac{x(\omega - \omega_0) + x(\omega + \omega_0)}{2}
 \end{aligned}$$

$$F\{\cos \omega_0 t\} = \pi [\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]$$



Examples

$$f_1(t) = e^{-at} u(t) \quad (a > 0)$$

$$= \int_0^{\infty} e^{-at} e^{-j\omega t} dt$$

$$= \int_0^{\infty} e^{-(j\omega + a)t} dt$$

$$= \frac{1}{-(j\omega + a)} e^{-(j\omega + a)t} \bigg|_0^{\infty} = \frac{1}{(j\omega + a)}$$

lim
 $a \rightarrow 0$ $\frac{1}{j\omega}$

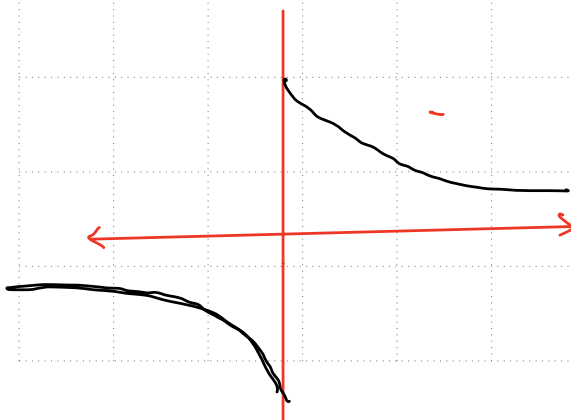
$$f(t) = \begin{cases} e^{-at} & t > 0 \\ -e^{at} & t < 0 \end{cases} = f_1(t) - f_2(t)$$

$$F(f(t)) = \frac{1}{j\omega + a} - \frac{1}{a - j\omega}$$

$$= \frac{-2j\omega}{\omega^2 + a^2}$$

When $a \rightarrow 0$ $\text{Sgn}(x)$

$$F(\text{sgn}(t)) = \frac{-2j}{\omega}$$

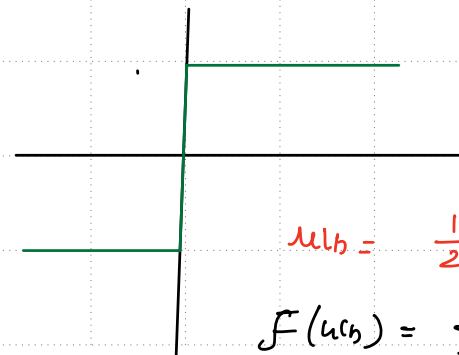


$$f_2(t) = e^{at} u(-t) \quad (a > 0)$$

$$F(f_2(t)) = \int_{-\infty}^0 e^{at} e^{-j\omega t} dt$$

$$= \frac{1}{a - j\omega}$$

$$F\left[\int_{-\infty}^t x(\tau) d\tau\right] = \frac{1}{j\omega} X(\omega) + \pi x(0) \delta(\omega)$$



$$u(t) = \frac{1}{2} [1 + \text{sgn}(t)]$$

$$F(u(t)) = \frac{1}{2} [2\pi \delta(\omega) + \frac{2}{j\omega}]$$

$$= \frac{1}{j\omega} + \pi \delta(\omega)$$